Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

Configuring AWS CloudWatch Connectors

The AWS CloudWatch connector allows Stellar Cyber to collect AWS ECS/EKS, WAF, VPC Flow, and Amazon GuardDuty logs. Use these instructions to obtain credentials and set up the connector.

Although the AWS CloudWatch connector may successfully import other logs, only AWS Elastic Container Service (ECS)/Elastic Kubernetes Service (EKS), Web Application Firewall (WAF), VPC Flow, and Amazon GuardDuty logs are supported in this release due to performance issues that can occur with other log types. Stellar Cyber is working on enhancements for other log types in future releases.

You can use an AWS CloudWatch connector to ingest logs from Amazon GuardDuty findings by exporting those logs to your AWS CloudWatch service. Refer to Amazon's documentation for configuration of this export: https://docs.aws.amazon.com/guardduty/latest/ug/guardduty_exportfindings.html.

Stellar Cyber connectors with the Collect function (collectors) may skip collecting some data when the ingestion volume is large, which potentially can lead to data loss. This can happen when the processing capacity of the collector is exceeded.

Connector Overview: AWS CloudWatch

Capabilities

- Collect: **Yes**
- Respond: **No**
- Native Alerts Mapped: **No**
- Runs on: **DP**
- Interval: **Configurable**

Collected Data

Content Type	Index	Locating Records

N/A	Syslog
	Traffic (VPC flow)
		msg_class:
		`aws_cloudwatch_log`
		`aws_cloudwatch_waf` (for WAF logs)
		`aws_guardduty` (for GuardDuty logs)
		`aws_cloudwatch_vpcflow` (for VPC Flow logs. Select the VPC Flow Logs checkmark also.)
		
		msg_origin.source:
		`aws_cloudwatch`
		`aws_guardduty`
		
		msg_origin.vendor:
		`aws`
		
		msg_origin.category:
		`paas`
		`waf` (for WAF logs)

Domain

https://logs.<Region>.amazonaws.com/

where <Region> is a variable from the configuration of this connector

Response Actions

N/A

Third Party Native Alert Integration Details

N/A

Required Credentials and Configurations

- For Access Key / Secret Key Auth Method: Access Key and Secret Key, plus Region, and Log Group(s)
- For IAM Role Auth Method: Source Account Access Key, Source Account Secret Key, and IAM Role ARN, plus Region, and Log Group(s)

Adding an AWS CloudWatch Connector

To add an AWS CloudWatch connector:

1. Add an AWS user
2. Create an access key

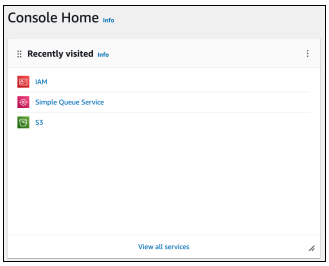
3. Identify CloudWatch log groups
4. (Optional) Create an IAM role in same AWS account
5. (Optional) Create an IAM role across AWS accounts
6. Add the connector in Stellar Cyber
7. Test the connector
8. Verify ingestion

Adding an AWS User for Stellar Cyber

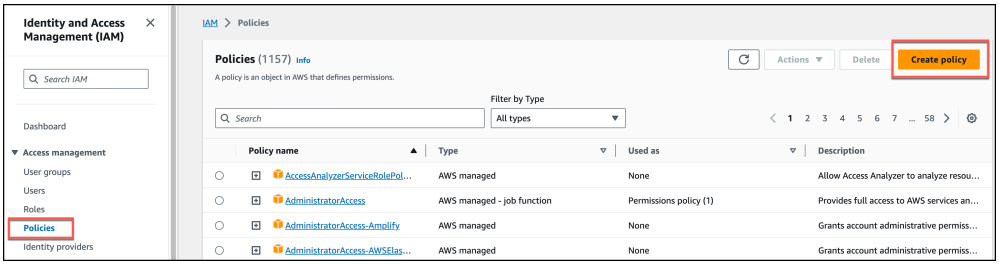
Use our example as a guideline, as you might be using a different software version.

To add a user with the appropriate permissions:

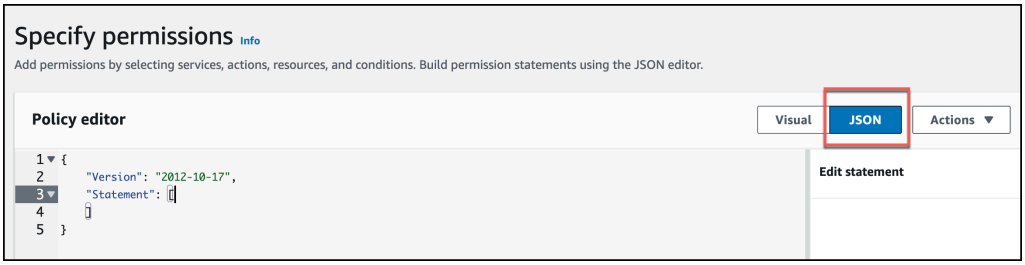
1. Log in to your **AWS Management Console** at https://aws.amazon.com/console. View the services in the Console Home or choose **View all services**.



2. Choose **IAM**. The **IAM Dashboard** appears.
3. Choose **Policies** and then choose **Create Policy**.



4. In the **Create policy** pane, choose the **JSON** tab.



5. Using the example below as a guide, edit the JSON policy document.

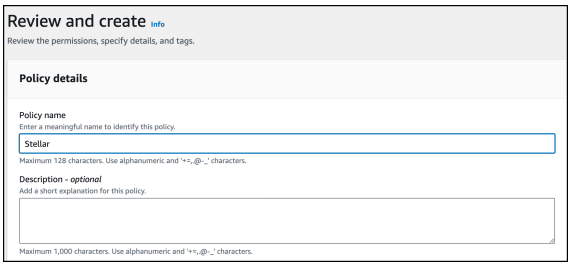
The following is just an EXAMPLE. You must modify this JSON to match the resources in your own environment.

CloudWatch policy

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
```

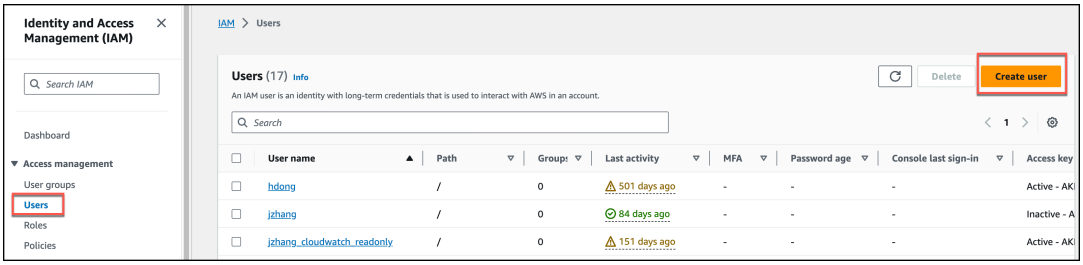
```
            "Effect": "Allow",
            "Action": [
                "logs:DescribeLogGroups",
                "logs:DescribeLogStreams",
                "logs:GetLogEvents",
                "logs:FilterLogEvents"
            ],
            "Resource": "*",
            "Condition": {
                "ForAnyValue:StringEquals": {
                    "aws:RequestedRegion": "us-west-2"
                }
            }
        }
    ]
}
```

6. Choose **Next**.
7. Give your policy a name to associate it with Stellar Cyber, then choose **Create policy**.

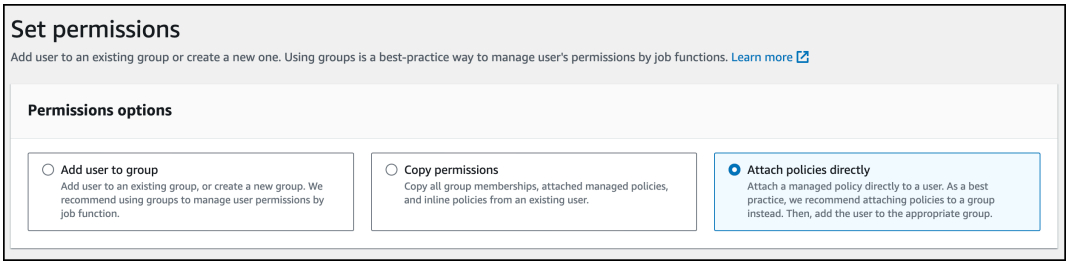


The policy can now be attached to a user.

8. From the IAM navigation pane, choose **Users** and then choose **Create user**.



9. On the **Specify user details** page for **User details**, enter a **User name** for the new user.
10. Choose **Provide user access to the – AWS Management Console** *optional* to produce login credentials for the new user, such as a password.
11. Choose how you want to create the Console password and then choose **Next**.
12. On the **Set permissions** page, choose **Attach policies directly**. Then search for the policy you created above.

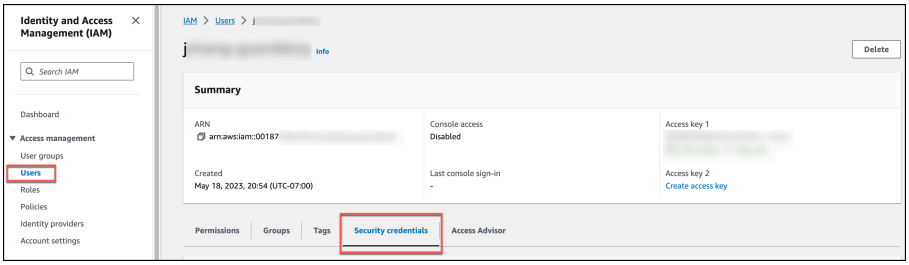


13. Select the checkbox to the left of the policy name, then choose **Next**.
14. On the **Review and create** page, verify the information and then choose **Create user**.

Creating an Access Key

The procedures in this section a or the Access Key / Secret Key Auth Method in Stellar Cyber.

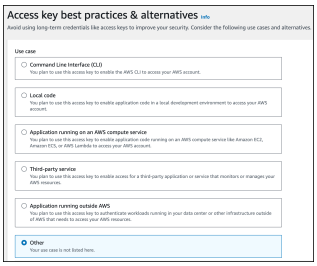
1. Log in to the IAM Console with your AWS account ID, your IAM user name, and password. You must have the permissions required to create access keys for a user.
2. Choose **Users**, then click the user name, and click **Security credentials**.



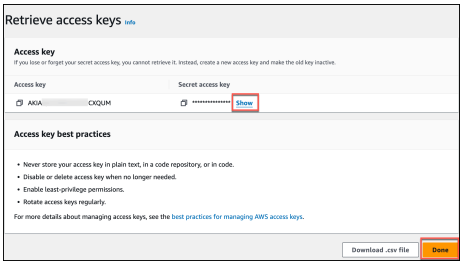
3. In the **Access keys** section, choose **Create access key**. Access keys have two parts: an access key and a secret access key. Click **Create access key**.



4. On the **Access key best practices & alternatives** page, choose **Other** and then choose **Next**.



5. (Optional) On the **Set description tag** page, enter a description.
6. Click **Create access key**.
7. On the **Retrieve access keys** page, choose **Show** to reveal the value of the user's secret access key. Save the access key and secret access key in a secure location. You will need them when configuring the connector in Stellar Cyber.

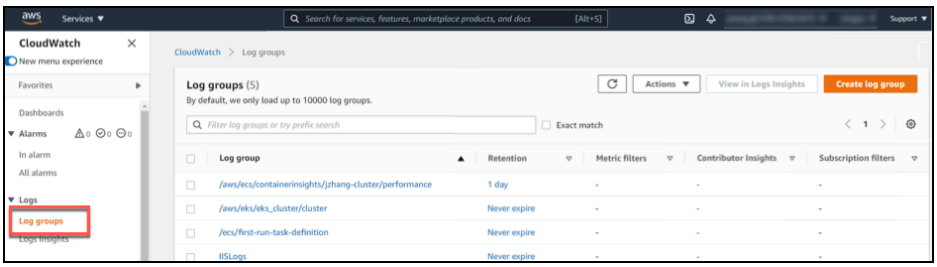


8. Click **Done**.

Identifying AWS CloudWatch Log Groups

For this connector, you will specify one or more log groups to ingest. Use this procedure to locate the group names (this procedure assumes that you have already configured your CloudWatch instances and log groups).

1. Log in to your **AWS CloudWatch Console** at https://console.aws.amazon.com/cloudwatch/.
2. Navigate to the **Logs** section of the left hand navigation pane.
3. Select **Log Groups**.



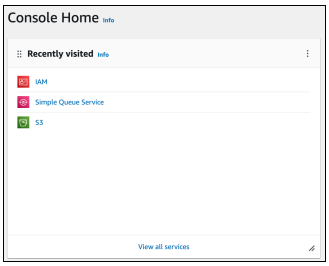
4. Make note of the groups you want to ingest to Stellar Cyber.

(Optional) Creating an IAM Role in Same AWS Account

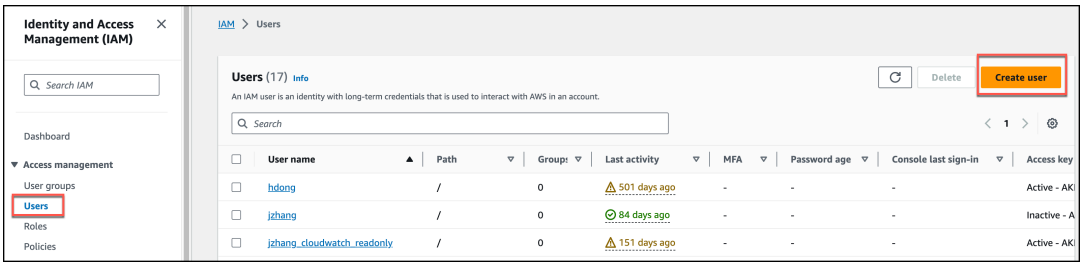
The procedures in this section are required for users who want to use the IAM Role Auth Method for the AWS resources, for an IAM role that is created in the same AWS account.

Create an IAM User Without Permissions and Create IAM Policy

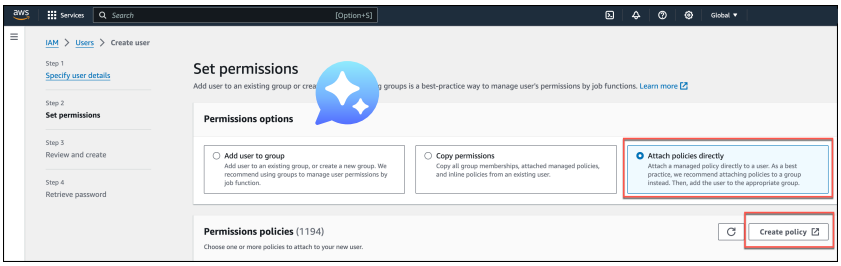
1. Log in to your **AWS Management Console** at https://aws.amazon.com/console. View the services in the Console Home or choose **View all services**.



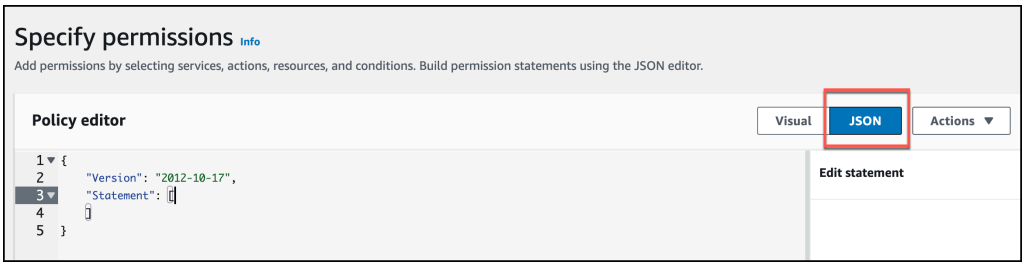
2. Choose **IAM**. The **IAM Dashboard** appears.
3. From the IAM navigation pane, choose **Users** and then click **Create user**.



4. On the **Specify user details** page for **User details**, enter a **User name** for the new user.
5. Choose **Provide user access to the – AWS Management Console** *optional* to produce login credentials for the new user, such as a password.
6. Choose how you want to create the Console password and then choose **Next**.
7. On the **Set permissions** page, choose **Attach policies directly** and click **Create policy**.



8. On the **Specify permissions** page, choose the **JSON** tab.



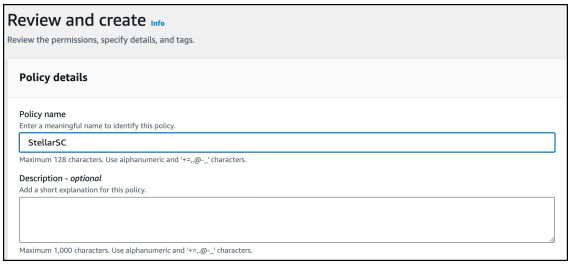
9. Using the example below as a guide, edit the JSON policy document.

> **!** The following is just an EXAMPLE. You must modify this JSON to match the resources in your own environment.

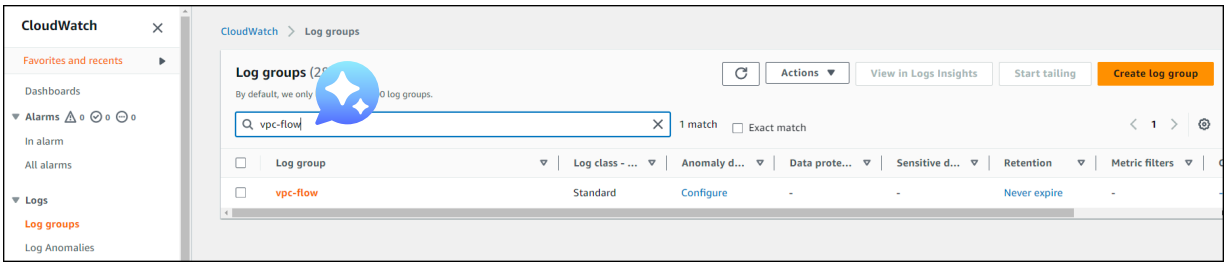
CloudWatch policy - IAM Role - same AWS account

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "logs:DescribeLogGroups",
                "logs:DescribeLogStreams",
                "logs:GetLogEvents",
                "logs:FilterLogEvents"
            ],
            "Resource": "arn:aws:logs:us-west-2:001xxxxxx433:log-
group:vpc-flow:*"
        }
    ]
}
```

10. Click **Next**.
11. Give your policy a name to associate it with Stellar Cyber, then choose **Create policy**.

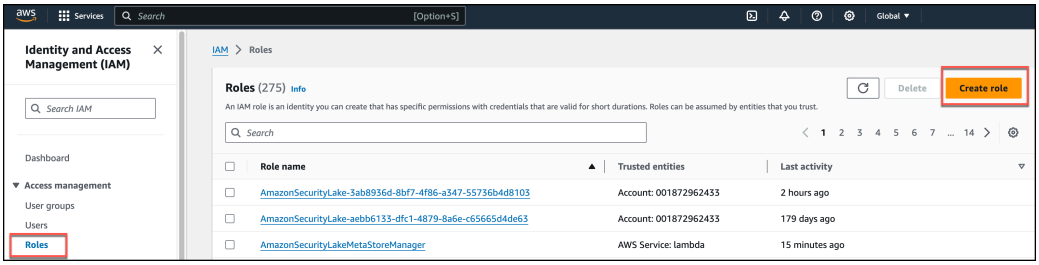


12. Note that in the example policy above, `vpc-flow` is the AWS CloudWatch log group name. To find your log group name, see Identifying AWS CloudWatch Log Groups.

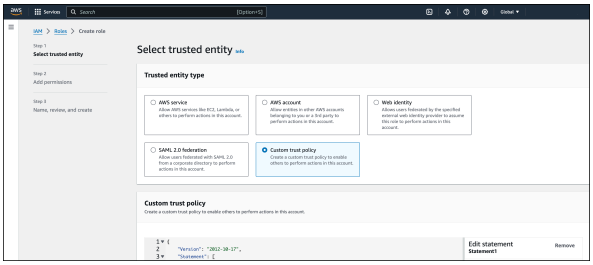


Create an IAM Role with Trusted Entities

1. From the IAM navigation pane, choose **Roles** and then click **Create Role**.



2. Choose **Custom trust policy**.



3. Using the example below as a guide, edit the custom trust policy document.

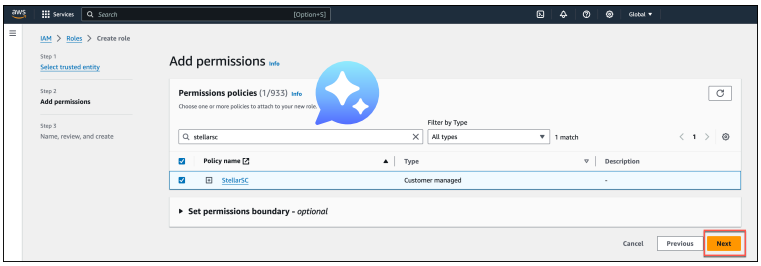
> **!** The following is just an EXAMPLE. You must modify this JSON to match the resources in your own environment.

Trusted entities - same AWS account

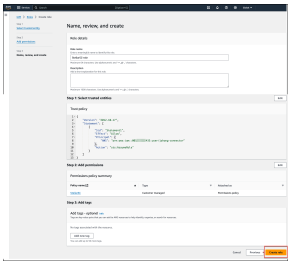
```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "Statement1",
            "Effect": "Allow",
            "Principal": {
                "AWS": "arn:aws:iam::001xxxxxx433:user/jzhang-
connector"
            },
            "Action": "sts:AssumeRole"
        }
    ]
}
```

4. Click **Next**.

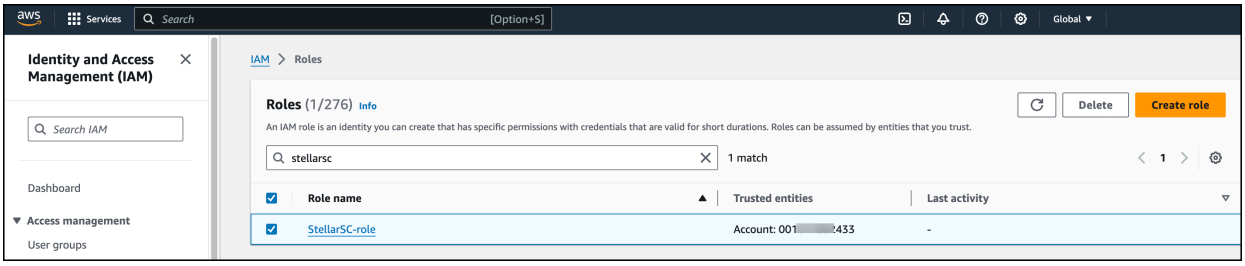
5. On the **Add permissions** page, select the permissions policy created earlier.



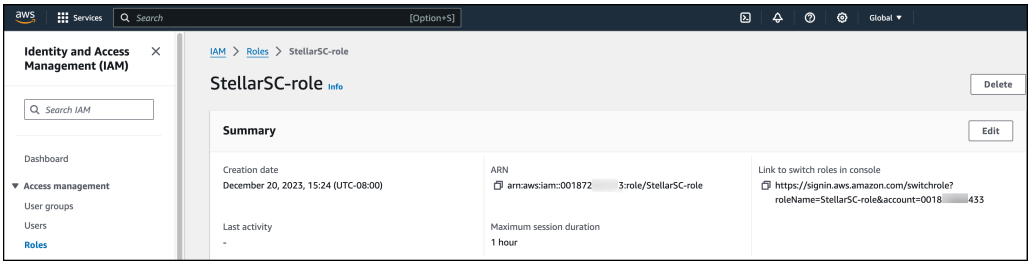
6. Click **Next**.
7. On the **Name, review, and create** page, enter a **Role name**.



8. Review the trusted entities and permissions, then click **Create Role**.
9. On the **Roles** page, use the Search box to find your role name.



10. Click your role name.



11. Copy your ARN, which you will need when configuring the connector in Stellar Cyber.

(Optional) Creating an IAM Role Across AWS Accounts

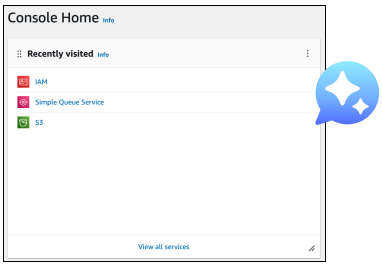
The procedures in this section are required for users who want to use the IAM Role Auth Method for the AWS resources, for an IAM role that is created across AWS accounts.

Create an IAM User for Source AWS Account and Create IAM Policy

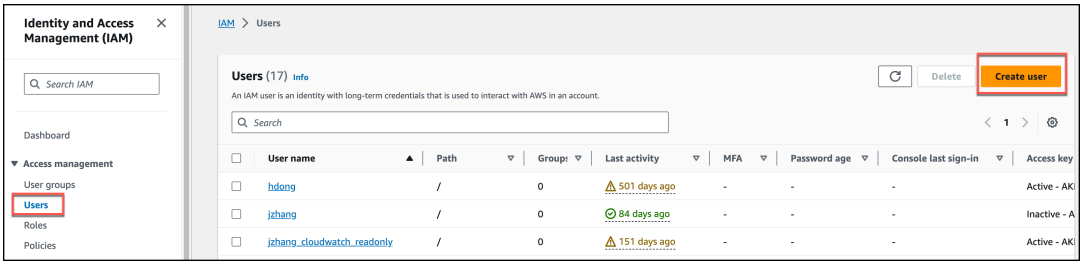
Create this IAM user if you have a source AWS account like an MSSP AWS account.

When configuring the connector in Stellar Cyber, you will need the source AWS account (MSSP account) and the access key and secret key.

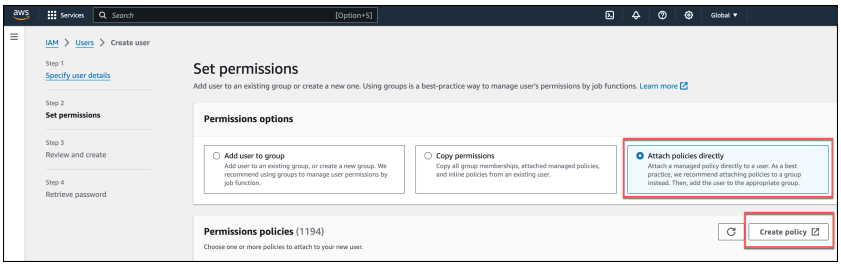
1. Log in to your **AWS Management Console** at https://aws.amazon.com/console. View the services in the Console Home or choose **View all services**.



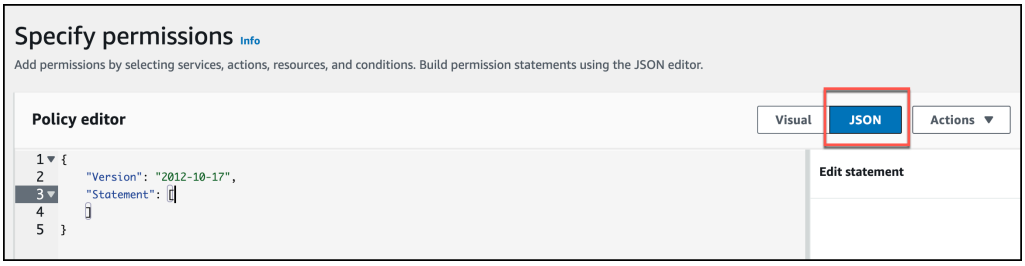
2. Choose **IAM**. The **IAM Dashboard** appears.
3. From the IAM navigation pane, choose **Users** and then click **Create user**.



4. On the **Specify user details** page, enter a **User name** for the new user.
5. Choose **Provide user access to the – AWS Management Console** *optional* to produce login credentials for the new user, such as a password.
6. Choose how you want to create the Console password and then choose **Next**.
7. Choose **Attach policies directly** and click **Create policy**.



8. On the **Specify permissions** page, choose the **JSON** tab.



9. Using the example below as a guide, edit the JSON policy document.

> The following is just an EXAMPLE. You must modify this JSON to match the resources in your own environment.

CloudWatch policy - IAM Role - across AWS accounts

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "sts:AssumeRole"
            ],
```

```
            "Resource": [
                "*"
            ]
        }
    ]
}
```

10. (Optional) To only allow the particular destination AWS account IAM role, using the example below as a guide, edit the JSON policy document.

    The following is just an EXAMPLE. You must modify this JSON to match the resources in your own environment.

    CloudWatch policy - IAM Role - allow destination AWS account

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "sts:AssumeRole"
            ],
            "Resource": [
                "arn:aws:iam::578xxxxxx273:role/Stellar-cloudwatch-
role"
            ]
        }
    ]
}
```

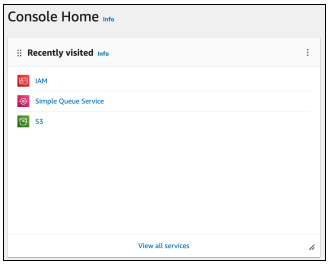
11. Click **Next**.
12. Give your policy a name to associate it with Stellar Cyber, then choose **Create policy**.
13. To find your log group name, see Identifying AWS CloudWatch Log Groups.

(Optional) Create an IAM User for Destination AWS Account and Create IAM Policy

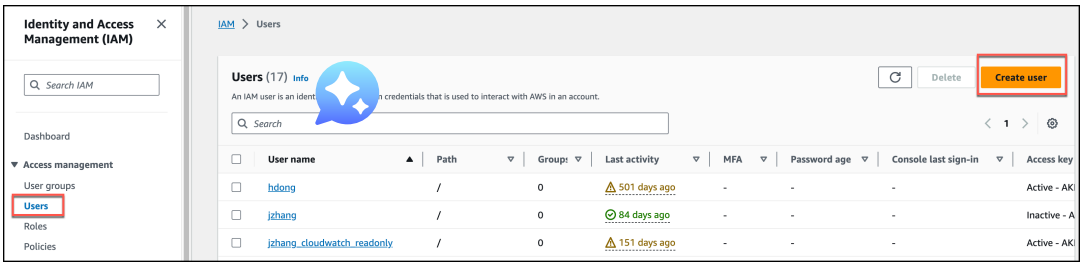
Create this IAM user if you have a destination AWS account like an MSSP's tenant AWS account.

When configuring the connector in Stellar Cyber, you will need the destination AWS account (MSSP's tenant account), the IAM role ARN, as well as the CloudWatch log group name.

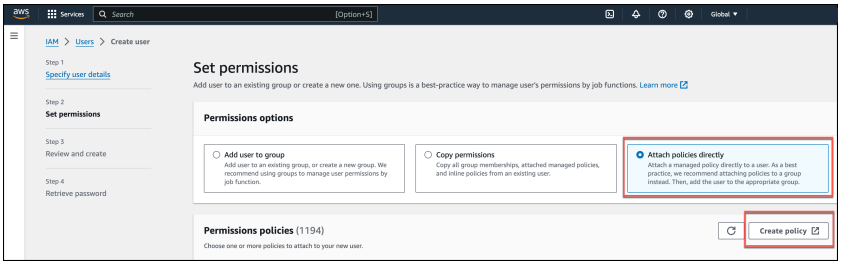
1. Log in to your **AWS Management Console** at https://aws.amazon.com/console. View the services in the Console Home or choose **View all services**.



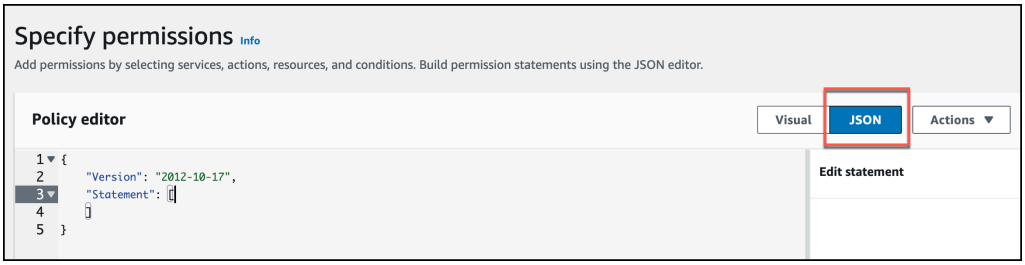
2. Choose **IAM**. The **IAM Dashboard** appears.
3. From the IAM navigation pane, choose **Users** and then click **Create user**.



4. On the **Specify user details** page, enter a **User name** for the new user.
5. Choose **Provide user access to the – AWS Management Console** *optional* to produce login credentials for the new user, such as a password.
6. Choose how you want to create the Console password and then choose **Next**.
7. Choose **Attach policies directly** and click **Create policy**.



8. On the **Specify permissions** page, choose the **JSON** tab.



9. Using the example below as a guide, edit the JSON policy document.

> The following is just an EXAMPLE. You must modify this JSON to match the resources in your own environment.

CloudWatch policy - Stellar-cloudwatch-policy

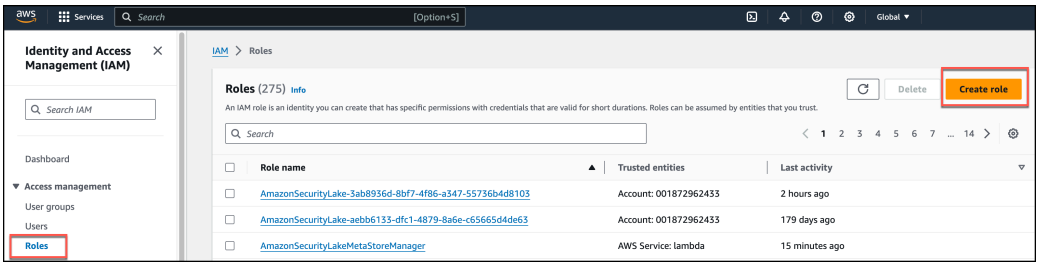
```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "logs:DescribeLogGroups",
                "logs:DescribeLogStreams",
                "logs:GetLogEvents",
                "logs:FilterLogEvents"
            ],
            "Resource": "arn:aws:logs:us-west-2:578xxxxx273:log-
group:System:*:*"
        }
    ]
}
```

10. Click **Next**.
11. Give your policy a name to associate it with Stellar Cyber, then choose **Create policy**.

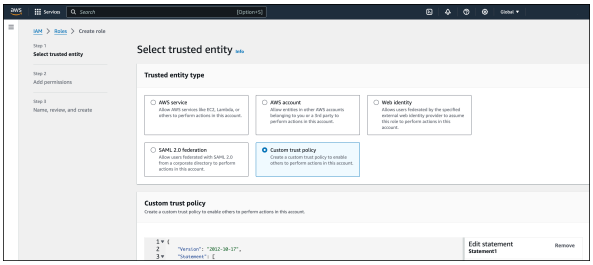
12. To find your log group name, see Identifying AWS CloudWatch Log Groups.

Create an IAM Role with Trusted Entities

1. From the IAM navigation pane, choose **Roles** and then click **Create Role**.



2. Choose **Custom trust policy**.



3. Using the example below as a guide, edit the custom trust policy document.

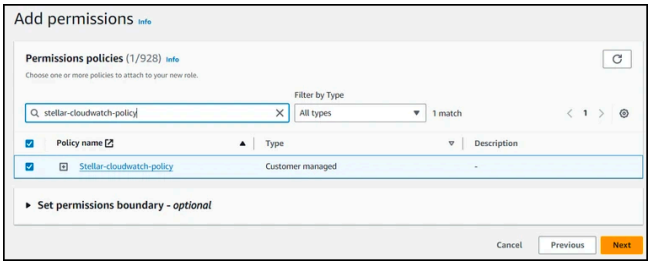
> The following is just an EXAMPLE. You must modify this JSON to match the resources in your own environment.

Trusted entities - across AWS accounts

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "Statement1",
            "Effect": "Allow",
            "Principal": {
                "AWS": "arn:aws:iam::001xxxxxx433:user/jzhang-
connector"
            },
            "Action": "sts:AssumeRole"
        }
    ]
}
```

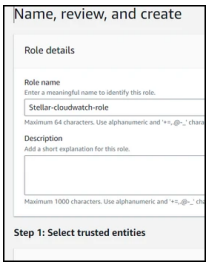
4. Click **Next**.

5. On the **Add permissions** page, select the permissions policy created earlier.



6. Click **Next**.

7. On the **Name, review, and create** page, enter a **Role name**.

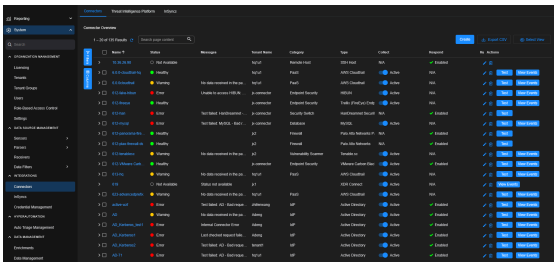


8. Review the trusted entities and permissions, then click **Create Role**.
9. On the **Roles** page, use the Search box to find your role name.
10. Click your role name.
11. Copy your ARN, which you will need when configuring the connector in Stellar Cyber.

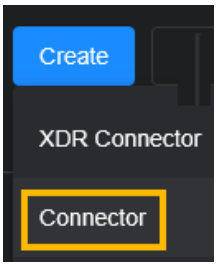
Adding the Connector in Stellar Cyber

With the credential and log information available, you can now add the AWS CloudWatch connector in Stellar Cyber:

1. Log in to Stellar Cyber.
2. Click **System | INTEGRATIONS | Connectors**. The **Connector Overview** appears.

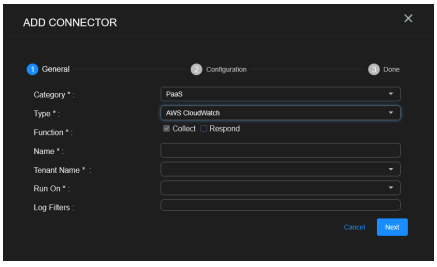


3. Click **Create** and select **Connector**.



The **General** tab of the **Add Connector** screen appears. The information on this tab cannot be changed after you add the connector.

4. Choose **PaaS** from the **Category** dropdown.
5. Choose **AWS CloudWatch** from the **Type** dropdown.

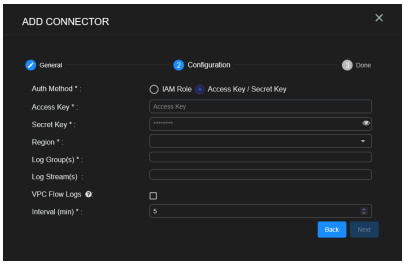


The asterisk (*) indicates a required field.

6. For this connector, the supported **Function** is **Collect**, which is enabled already.
7. Enter a **Name**.

   Notes:
   - This field does not accept multibyte characters.
   - It is recommended that you follow a naming convention such as tenantname-connectortype.

8. Choose a **Tenant Name**. The Interflow records created by this connector include this tenant name.
9. Choose the device on which to run the connector.
10. (Optional) When the **Function** is **Collect**, you can apply **Log Filters**. For information, see Managing Log Filters.
11. Click **Next**. The **Configuration** tab appears.



   The asterisk (*) indicates a required field.

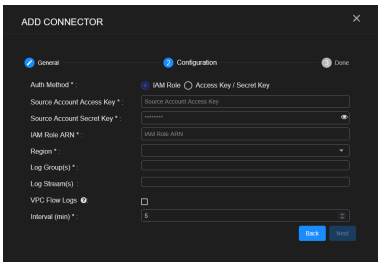
12. Choose the **Auth Method** to use **Access Key / Secret Key** or **IAM Role**.

    For **Access Key / Secret Key** Auth Method:

    a. Enter the **Access Key** you noted above in Creating an Access Key.

    b. Enter the **Secret Key** you noted above.

    For **IAM Role** Auth Method:



    The asterisk (*) indicates a required field.

    a. Enter the **Source Account Access Key**.

    b. Enter the **Source Account Secret Key**.

    The IAM role can be across multiple AWS accounts:
    - For the source account, provide the trusted IAM user's access key and secret key. Refer to Creating an Access Key.
    - For the destination account, provide the AWS IAM role, in which this role is trusting the source account IAM user. Refer to

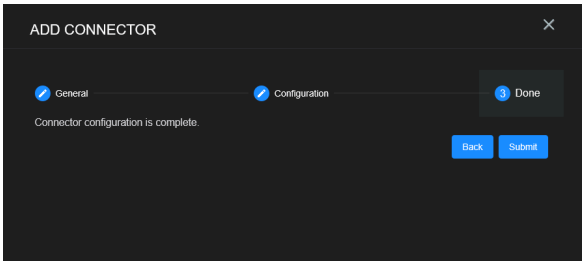
(Optional) Creating an IAM Role Across AWS Accounts.

c. Enter the **IAM Role ARN** you noted above.

13. Choose a **Region** from the available regions in the dropdown.
14. Enter the **Log Group(s)** as named in your AWS CloudWatch console. To enter multiple log group names, press **Enter** after each name.

    If you want to enter **Log Stream(s)** in the optional step below, you can only enter one Log Group in this step.

15. (Optional) Enter the **Log Stream(s)**. Provide a list of log stream(s) that correspond to the provided log group. To enter multiple log stream names, press **Enter** after each name. If log streams are not specifically listed, every log stream within the provided log group is automatically searched.
16. (Optional) Click the checkmark for **VPC Flow Logs** if you are collecting VPC flow logs.
17. Choose the **Interval (min)**. This is how often the logs are collected.
18. Click **Next**. The final confirmation tab appears.



19. Click **Submit**.

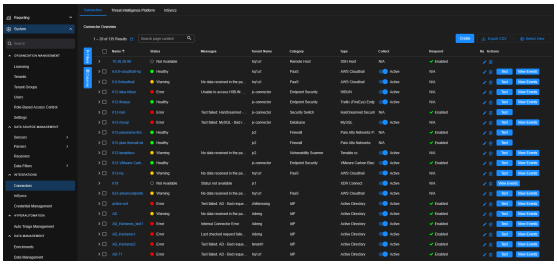
The new connector is immediately active and starts collecting logs.

Testing the Connector

When you add (or edit) a connector, we recommend that you run a test to validate the connectivity parameters you entered. The test validates authentication and connectivity.

For connectors running on a sensor, Stellar Cyber recommends that you allow 30-60 seconds for new or modified configuration details to be propagated to the sensor before performing a test.

1. Click **System | INTEGRATIONS | Connectors**. The **Connector Overview** appears.



2. Locate the connector by name that you added, or modified, or that you want to test.
3. Click **Test** at the right side of that row. The test runs immediately.

Note that you may run only one test at a time.

Stellar Cyber conducts a basic connectivity test for the connector and reports a success or failure result. A successful test indicates that you entered all of the connector information correctly.

To aid troubleshooting your connector, the dialog remains open until you explicitly close it by using the **X** button. If the test fails, you can select the button from the same row to review and correct issues.

The connector status is updated every five (5) minutes. A successful test clears the connector status, but if issues persist, the status reverts to failed after a minute.

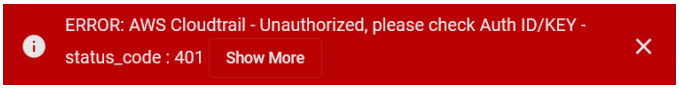
Repeat the test as needed.

Display sample messages...

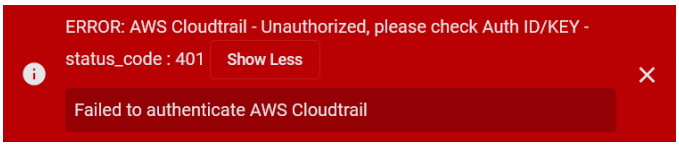
Success !



Failure with summary of issue:



Show More example detail:



If the test fails, the common HTTP status error codes are as follows:

HTTP Error Code	HTTP Standard Error Name	Explanation	Recommendation
400	Bad Request	This error occurs when there is an error in the connector configuration.	Did you configure the connector correctly?
401	Unauthorized	This error occurs when an authentication credential is invalid or when a user does not have sufficient privileges to access a specific API.	Did you enter your credentials correctly? Are your credentials expired? Are your credentials entitled or licensed for that specific resource?
403	Forbidden	This error occurs when the permission or scope is not correct in a valid credential.	Did you enter your credentials correctly? Do you have the required role or permissions for that credential?
404	Not Found	This error occurs when a URL path does not resolve to an entity.	Did you enter your API URL correctly?
429	Too Many Requests	This error occurs when the API server receives too much traffic or if a user's license or entitlement quota is exceeded.	The server or user license/quota will eventually recover. The connector will periodically retry the query.

HTTP Error Code	HTTP Standard Error Name	Explanation	Recommendation
			If this occurs unexpectedly or too often, work with your API provider to investigate the server limits, user licensing, or quotas.

For a full list of codes, refer to HTTP response status codes.

Verifying Ingestion

To verify ingestion:

1. Select **Threat Hunting**.

   The **Interflow Search** tab appears.

2. Change the **Indices** for the type of content you collected:

   - Change the **Indices** to **Syslog**.
   - For VPC flow logs only, change the **Indices** to **Traffic**.

   The table immediately updates to show ingested Interflow records.

© 2026 Stellar Cyber . All rights reserved.